

Testing in Linear Models

5 Mar, 2026

$$\underline{y} = X \underline{\theta} + \underline{\varepsilon}, \quad \underline{\varepsilon} \sim N_n(\underline{0}, \sigma^2 I)$$

$$\hat{\underline{\theta}}_{LS} = (X^T X)^{-1} X^T \underline{y} \sim N_d(\underline{\theta}, \sigma^2 (X^T X)^{-1}) \quad \text{indep.}$$

$$\underline{e} \sim N_n(\underline{0}, \sigma^2 (I - H))$$

$$\|\underline{e}\|^2 = \sum_{i=1}^n e_i^2 \sim \sigma^2 \chi_{n-d}^2 \quad \text{independently of } \hat{\underline{\theta}}_{LS}$$

$$H_0: \underline{a}^T \underline{\theta} = 0 \quad \text{vs.} \quad H_a: \underline{a}^T \underline{\theta} \neq 0$$

$$\underline{a}^T \hat{\underline{\theta}}_{LS} \sim N(\underline{a}^T \underline{\theta}, \sigma^2 \underline{a}^T (X^T X)^{-1} \underline{a}) \quad \text{indep.}$$

$$\|\underline{e}\|^2 \sim \sigma^2 \chi_{n-d}^2$$

- $Z \sim N(\mu, \sigma^2)$, σ^2 unknown

$$W \sim \sigma^2 \chi_k^2$$

Then, $\frac{Z - \mu}{\sigma} \sim N(0, 1)$

$$\frac{W}{\sigma^2} \sim \chi_k^2$$

$$\frac{Z - \mu}{\sigma} / \sqrt{\frac{W}{k\sigma^2}} \sim t_k$$

$$= \frac{Z - \mu}{\sqrt{\frac{W}{k}}}$$

$$T := \frac{Z}{\sqrt{\frac{W}{k}}}$$

Then, under $\mu = 0$,

$$T \sim t_k$$

The dist. of T is known under $H_0: \mu = 0$

So, T can be used for testing
 $H_0: \mu = 0$ vs. $H_a: \mu \neq 0$.

$$E(T) = E\left(\frac{Z}{\sigma}\right) E\left(\frac{1}{\sqrt{\frac{n}{k\sigma^2}}}\right)$$

$$= \mu \cdot c \quad \text{where } c = E\left(\frac{1}{\sqrt{\frac{\sum x_k^2}{n}}}\right)$$

So, we reject $H_0: \mu = 0$ if

- T is either very large or very small

($\Rightarrow$) $|T|$ is large (due to symmetry of t dist)

$H_a: \mu \neq 0$

- T is large

$H_a: \mu > 0$

- T is small

$H_a: \mu < 0$

$$H_0: \underline{a}^T \underline{\theta} = 0 \quad \text{vs.} \quad H_a: \underline{a}^T \underline{\theta} \neq 0$$

$$\underline{a}^T \hat{\underline{\theta}}_{LS} \sim N(\underline{a}^T \underline{\theta}, \sigma^2 \underline{a}^T (X^T X)^{-1} \underline{a})$$

$$\Rightarrow \frac{\underline{a}^T \hat{\underline{\theta}}_{LS}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}}} \sim N\left(\frac{\underline{a}^T \underline{\theta}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}}}, \sigma^2\right) \xrightarrow{\text{indep}}$$

$$\hat{\sigma}^2 = \frac{1}{n-d} \sum_{i=1}^n e_i^2 \quad \text{satisfies} \quad (n-d) \hat{\sigma}^2 \sim \sigma^2 \chi_{n-d}^2$$

$$\therefore T = \frac{\underline{a}^T \hat{\underline{\theta}}_{LS}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}}} \cdot \frac{1}{\sqrt{\hat{\sigma}^2}} \sim t_{n-d} \quad \text{under } H_0$$

$$\therefore T = \frac{\underline{a}^T \hat{\underline{\theta}}_{LS}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a} \hat{\sigma}^2}} \sim t_{n-d} \text{ under } H_0.$$

We reject H_0 at level α if

- For $H_a: \underline{a}^T \underline{\theta} \neq 0$, $|T| > t_{\alpha/2; n-d}$

- For $H_a: \underline{a}^T \underline{\theta} > 0$, $T > t_{\alpha; n-d}$

- For $H_a: \underline{a}^T \underline{\theta} < 0$, $T < -t_{\alpha; n-d}$

Focus on $H_a: \mu \neq 0$.

Rejecting for large values of $|T|$

$\Updownarrow$
Rejecting for large values of T^2

$T \sim t_{n-d}$ under $H_0 \Leftrightarrow T^2 \sim F_{1, n-d}$ under H_0

Rejecting if $|T| > t_{\alpha/2; n-d}$

$\Updownarrow$
Rejecting if $F = T^2 > F_{\alpha; 1, n-d}$

$$F = T^2 = \left[\frac{\underline{a}^T \hat{\underline{\theta}}_{LS}}{\sqrt{\hat{\sigma}^2 \underline{a}^T (X^T X)^{-1} \underline{a}}} \right]$$

$$= \frac{(\underline{a}^T \hat{\underline{\theta}}_{LS})^2}{\hat{\sigma}^2 \underline{a}^T (X^T X)^{-1} \underline{a}} \sim F_{1, n-d} \quad \text{under } H_0: \underline{a}^T \underline{\theta} = 0$$

Reject H_0 if $F > F_{\alpha; 1, n-d}$.

Now, consider

$$H_0: \underline{\theta} = \underline{0} \quad \text{vs.} \quad H_a: \underline{\theta} \neq \underline{0}.$$

[To check whether our model is at all needed]

$$\hat{\underline{\theta}}_{LS} \sim N_d(\underline{\theta}, \sigma^2 (X^T X)^{-1}) \quad \text{independent.}$$

$$(n-d) \hat{\sigma}^2 \sim \sigma^2 \chi^2_{n-d}$$

$$\begin{aligned} \underline{Z} &\sim N(\underline{\mu}, \sigma^2 V) \\ \underline{W} &\sim \sigma^2 \chi^2_k \end{aligned} \quad \text{indep.}$$

V is known

Definition: If $\underline{z} \sim N_p(\underline{0}, I)$, then $\|\underline{z}\|^2 \sim \chi_p^2$

• If $\underline{z} \sim N_p(\underline{\mu}, I)$, then $\|\underline{z}\|^2 \sim \chi_p^2(\delta)$

$\delta = \|\underline{\mu}\|^2$ is the noncentrality
non-central χ^2 dist. parameter

• If $\underline{z} \sim N_p(\underline{0}, \Sigma)$, with Σ p.d., then
 Σ has a p.d. square root $\Sigma^{1/2}$ ($\Sigma^{1/2} \Sigma^{1/2} = \Sigma$)
 $\Sigma^{1/2}$ is invertible (i.e., $\Sigma^{-1/2}$ exists)

$$\Sigma^{-1/2} \underline{z} \sim N_p(\underline{0}, \underbrace{\Sigma^{-1/2} \Sigma \Sigma^{-1/2}}_I)$$

$$\Rightarrow \|\Sigma^{-1/2} \underline{z}\|^2 \sim \chi_p^2, \quad \|\Sigma^{-1/2} \underline{z}\|^2 = \underline{z}^T \Sigma^{-1} \underline{z}$$

If $\underline{z} \sim N_p(\underline{0}, \Sigma)$, then $\underline{z}^T \Sigma^{-1} \underline{z} \sim \chi_p^2$.

• If $\underline{z} \sim N_p(\underline{\mu}, \Sigma)$, then $\underline{z}^T \Sigma^{-1} \underline{z} \sim \chi_p^2(\delta)$
 $\delta = \underline{\mu}^T \Sigma^{-1} \underline{\mu}$

• If $\underline{z} \sim N(\underline{0}, I)$ and P is an orthogonal projection matrix with $\text{rank}(P) = r$, then

$$\underline{z}^T P \underline{z} \sim \chi_r^2$$

P is an orthogonal projection with $\text{rank}(P) = r$

$\Rightarrow P = \sum_{i=1}^r \underline{u}_i \underline{u}_i^T$, where $\underline{u}_1, \dots, \underline{u}_r$ are
orthonormal vectors.

$$\underline{z}^T P \underline{z} = \underline{z}^T \sum_{i=1}^r \underline{u}_i \underline{u}_i^T \underline{z} = \sum_{i=1}^r (\underline{u}_i^T \underline{z})^2$$

$$\begin{pmatrix} \underline{u}_1^T \underline{z} \\ \vdots \\ \underline{u}_r^T \underline{z} \end{pmatrix} = \begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_r^T \end{pmatrix} \underline{z} \sim N_r \left(\begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_r^T \end{pmatrix} \underline{0}, \begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_r^T \end{pmatrix} I (\underline{u}_1 \dots \underline{u}_r) \right)$$

$$= N_r \left(\underline{0}, \underbrace{((\underline{u}_i^T \underline{u}_j))}_{\delta_{ij}} \right)$$

$$\delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$= N_r(\underline{0}, I)$$

$$\Rightarrow \sum_{i=1}^r (\underline{u}_i^T \underline{z})^2 \underset{\underline{z}^T P \underline{z}}{\sim} \chi_r^2$$

• If $\underline{z} \sim N_p(\underline{0}, I)$ and P is an orthogonal projection matrix, then $\underline{z}^T P \underline{z} \sim \chi_r^2$, where $r = \text{rank}(P) = \text{trace}(P) = \# \text{ of non zero eigenvalues of } P$.

• If $\underline{z} \sim N_p(\underline{\mu}, I)$ and P is an orthogonal projection matrix, then $\underline{z}^T P \underline{z} \sim \chi_r^2(\delta)$, where $r = \text{rank}(P) = \text{trace}(P)$, $\delta = \underline{\mu}^T P \underline{\mu}$

$\underline{z} \sim N(\underline{0}, \Sigma)$ Σ is p.d.

$\underline{z}^T A \underline{z}$ A is p.s.d.

$\underline{v} = \Sigma^{-1/2} \underline{z} \sim N(\underline{0}, I)$

$\underline{v} = \Sigma^{-1/2} \underline{z}$

$\underline{z} \sim \Sigma^{1/2} \underline{v}$, $\underline{v} \sim N(\underline{0}, I)$

$$\begin{aligned} \underline{z}^T A \underline{z} &= (\Sigma^{1/2} \underline{v})^T A (\Sigma^{1/2} \underline{v}) \\ &= \underline{v}^T \Sigma^{1/2} A \Sigma^{1/2} \underline{v} \quad \underline{v} \sim N(\underline{0}, \underline{I}) \end{aligned}$$

If $\Sigma^{1/2} A \Sigma^{1/2}$ is an orthogonal proj. matrix

$$\text{then } \underline{v}^T \Sigma^{1/2} A \Sigma^{1/2} \underline{v} \sim \chi^2_r,$$

$$\text{where } r = \text{rank}(\Sigma^{1/2} A \Sigma^{1/2}) = \text{trace}(\Sigma^{1/2} A \Sigma^{1/2}) = \text{trace}(A \Sigma)$$

$\Sigma^{1/2} A \Sigma^{1/2}$ is p.s.d. ($\because A$ is p.s.d. & $\Sigma^{1/2}$ is sym.)

$\uparrow$ symmetric

$\Sigma^{1/2} A \Sigma^{1/2}$ is an orthogonal projection if it is idempotent.

$$\text{i.e., } \Sigma^{1/2} A \Sigma^{1/2} \Sigma^{1/2} A \Sigma^{1/2} = \Sigma^{1/2} A \Sigma^{1/2} \quad (\Rightarrow) \quad A \Sigma A = A$$

- If $\underline{z} \sim N(\underline{0}, \Sigma)$ with Σ p.d. and A is a p.s.d. matrix. Then, $\underline{z}^T A \underline{z} \sim \chi_r^2$ if $A \Sigma A = A$.
In this case, $r = \text{trace}(A \Sigma)$.

In particular, taking $A = \Sigma^{-1}$, we get that
 $\underline{z}^T \Sigma^{-1} \underline{z} \sim \chi_p^2$.

- If $\underline{z} \sim N_p(\underline{\mu}, \Sigma)$ with Σ p.d. and A is a p.s.d. matrix. Then, $\underline{z}^T A \underline{z} \sim \chi_r^2(\delta)$ if $A \Sigma A = A$. In this case, $r = \text{trace}(A \Sigma)$ and $\delta = \underline{\mu}^T A \underline{\mu}$.

9 Mar 2026

$$\underline{y} = X \underline{\theta} + \underline{\varepsilon}, \quad \underline{\varepsilon} \sim N(\underline{0}, \sigma^2 \underline{I})$$

To test $H_0: \underline{\theta} = \underline{0}$ vs. $H_a: \underline{\theta} \neq \underline{0}$.

$$\bullet \hat{\underline{\theta}}_{LS} = (X^T X)^{-1} X^T \underline{y} \sim N(\underline{\theta}, \sigma^2 (X^T X)^{-1})$$

$$\bullet (n-d) \hat{\sigma}^2 = \|\hat{\underline{e}}\|^2 \sim \sigma^2 \chi_{n-d}^2$$

$\bullet \hat{\underline{\theta}}_{LS}$ & $\hat{\sigma}^2$ are independent.

$$\text{Under } H_0, \hat{\underline{\theta}}_{LS} \sim N(\underline{0}, \sigma^2 (X^T X)^{-1})$$

$$\underline{z} \sim N_p(\underline{\mu}, \Sigma), \quad \text{Then } (\underline{z} - \underline{\mu})^T \Sigma^{-1} (\underline{z} - \underline{\mu}) \sim \chi_p^2$$

$$\underline{z} \sim N_p(\underline{\mu}, \underline{I}), \quad A \text{ is an orth. proj. matrix, i.e. } (\underline{z} - \underline{\mu})^T A (\underline{z} - \underline{\mu}) \sim \chi_r^2, \quad r = \text{tr}(A) = \text{rank}(A)$$

$$\underline{z} \sim N_p(\underline{\mu}, \underline{\Sigma}), \quad \underline{z}^T \underline{\Sigma}^{-1} \underline{z} \sim \chi_p^2(\delta)$$

$$\delta = \underline{\mu}^T \underline{\Sigma}^{-1} \underline{\mu}$$

$$\underline{z} \sim N_p(\underline{\mu}, I), \quad A \text{ is orth. proj., lin.}$$

$$\underline{z}^T A \underline{z} \sim \chi_r^2(\delta), \quad r = \text{tr}(A) = \text{rank}(A)$$

$$\delta = \underline{\mu}^T A \underline{\mu}$$

$$\hat{\underline{\theta}}_{LS} \sim N_d(\underline{\theta}, \sigma^2 (X^T X)^{-1})$$

$$\hat{\underline{\theta}}_{LS}^T (\sigma^2 (X^T X)^{-1})^{-1} \hat{\underline{\theta}}_{LS} \sim \chi_d^2(\delta)$$

$$\delta = \underline{\theta}^T (\sigma^2 (X^T X)^{-1})^{-1} \underline{\theta}$$

$$\frac{\hat{\underline{\theta}}_{LS}^T X^T X \hat{\underline{\theta}}_{LS}}{\sigma^2}$$

$$= \frac{\underline{\theta}^T X^T X \underline{\theta}}{\sigma^2}$$

$$\frac{\hat{\theta}_{LS}^T X^T X \hat{\theta}_{LS}}{\sigma^2} \sim \chi_d^2(\delta), \quad \delta = \frac{\underline{\theta}^T X^T X \underline{\theta}}{\sigma^2}$$

$X^T X$ is p.d. ($\because X$ has full col. rank)

$$\Rightarrow \delta \geq 0, "=" \text{ iff } \underline{\theta} = \underline{0}.$$

$$(n-d) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-d}^2, \text{ indep. of } \hat{\theta}_{LS} \quad \Updownarrow H_0 \text{ holds.}$$

- If $U \sim \chi_m^2, V \sim \chi_n^2$ and they are indep.
 then $\frac{U/m}{V/n} \sim F_{m,n}$ (F dist with m, n d.f.)
- If $U \sim \chi_m^2(\delta), V \sim \chi_n^2$ and they are indep.
 then $\frac{U/m}{V/n} \sim F_{m,n}(\delta)$ (non-central F dist with (m, n) df and ncp δ)

$$\frac{\hat{\theta}_{LS}^T X^T X \hat{\theta}_{LS}}{\sigma^2} \sim \chi_d^2(\delta), \quad \delta = \frac{\theta^T X^T X \theta}{\sigma^2}$$

$$(n-1) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-d}^2 \quad \left. \vphantom{\frac{\hat{\sigma}^2}{\sigma^2}} \right\} \text{indep.}$$

$$\Rightarrow F = \frac{\hat{\theta}_{LS}^T X^T X \hat{\theta}_{LS} / d}{\hat{\sigma}^2} \sim F_{d, n-d}(\delta)$$

$$\delta \geq 0, \quad "=" \text{ iff } \underline{\theta} = \underline{0}$$

$$F \sim F_{d, n-d}(\delta), \quad \delta \geq 0, \quad "=" \text{ iff } H_0 \text{ holds.}$$

We expect to observe large values of F compared to $F_{d, n-d}$ (central F dist) if H_0 does not hold. So, we reject H_0 if the value of F is large.

F is our test statistic

We will reject if F is large (i.e., $F > c$)

$F \sim F_{d, n-d}$ under H_0 .

So, we reject at level α if

$F > F_{\alpha; d, n-d} \leftarrow$ upper α -quantile of $F_{d, n-d}$.

$$\underline{y} = X \underline{\hat{\theta}}_{LS} + \underline{\hat{e}}$$

$$\underbrace{\underline{y}^T \underline{y}}_{TSS} = \underbrace{\underline{\hat{\theta}}_{LS}^T X^T X \underline{\hat{\theta}}_{LS}}_{SSE} + \underbrace{\underline{\hat{e}}^T \underline{\hat{e}}}_{SSE}$$

$$SSE = \underline{y}^T \underline{y} - \underline{\hat{\theta}}_{LS}^T X^T X \underline{\hat{\theta}}_{LS}$$

Model under $H_0: \underline{\theta} = \underline{0}$

$$\underline{y} = \underline{0} + \underline{\varepsilon} \Rightarrow \underline{y} = \underline{\varepsilon}$$

In this case, $\hat{\underline{y}}_{H_0} = \underline{0} \Rightarrow \hat{\underline{e}}_{H_0} = \underline{y}$.

$$\Rightarrow SSE_{H_0} = \hat{\underline{e}}_{H_0}^T \hat{\underline{e}}_{H_0} = \underline{y}^T \underline{y} = TSS$$

$$SSE = \underline{y}^T \underline{y} - \hat{\underline{\theta}}_{LS}^T \underline{X}^T \underline{X} \hat{\underline{\theta}}_{LS}$$

$$SSE_{H_0} = \underline{y}^T \underline{y}$$

$$\therefore \hat{\underline{\theta}}_{LS}^T \underline{X}^T \underline{X} \hat{\underline{\theta}}_{LS} = SSE_{H_0} - SSE = SS(H_0)$$

$$F = \frac{\hat{\underline{\theta}}_{LS}^T \underline{X}^T \underline{X} \hat{\underline{\theta}}_{LS} / d}{\hat{\sigma}^2}$$

$$F = \frac{\hat{\theta}_{LS}^T X^T X \hat{\theta}_{LS} / d}{\hat{\sigma}^2}$$

$$\hat{\theta}_{LS}^T X^T X \hat{\theta}_{LS} = SS(H_0) = SSE_{H_0} - SSE$$

$$(n-d) \hat{\sigma}^2 = \|\hat{e}\|^2 = SSE$$

$$\therefore F = \frac{SS(H_0) / d}{SSE / (n-d)} = \frac{SS(H_0) / df(H_0)}{SSE / df(SSE)}$$

$$\underline{y}^T \underline{y} = \underline{y}^T H \underline{y} + \underbrace{\underline{y}^T (I-H) \underline{y}}_{SSE}$$

$$SSE \sim \sigma^2 \chi_{n-d}^2$$

$$\therefore df(SSE) = n-d$$

$$\text{Under } H_0 \quad df(SS E_{H_0}) = n$$

$$df(H_0) = df(SS E_{H_0}) - df(SSE) = n - (n-d) = d$$

The test statistic can be computed in the following way:

- Fit the full model: $\underline{y} = X\underline{\theta} + \underline{\varepsilon}$
Get $SSE = \|\underline{y} - X\hat{\underline{\theta}}_{LS}\|^2 = \|\hat{\underline{e}}\|^2$

- Fit the model restricted to $H_0: \underline{\theta} = \underline{0}$.

Get $SSE_{H_0} = \min_{\underline{\theta}: H_0 \text{ holds}} \|\underline{y} - X\underline{\theta}\|^2$

- Find $SS(H_0) = SSE_{H_0} - SSE$.

- Calculate $F = \frac{SS(H_0) / df(H_0)}{SSE / df(SSE)}$

where $df(SSE)$ is the df. of SSE

$$df(H_0) = df(SSE_{H_0}) - df(SSE).$$

We reject H_0 at level α if

$$F > F_{\alpha; df(H_0), df(SSE)}.$$

Our model: $\underline{y} = X\underline{\theta} + \underline{\varepsilon}$

X is non-random, $\underline{\varepsilon} \sim N(\underline{0}, \sigma^2 I)$.

Thus, $\underline{y} \sim N_n(X\underline{\theta}, \sigma^2 I)$, $\underline{\theta}, \sigma^2$ are unknown.

Likelihood of $\underline{\theta}, \sigma^2$

$$L(\underline{\theta}, \sigma^2) = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \|\underline{y} - X\underline{\theta}\|^2} \quad \left[\begin{array}{l} \underline{z} \sim N_d(\underline{\mu}, \Sigma) \\ L(\underline{\mu}, \Sigma) \\ = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (\underline{z} - \underline{\mu})^T \Sigma^{-1} (\underline{z} - \underline{\mu})} \end{array} \right]$$

$$L(\underline{\theta}, \sigma^2) = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \|\underline{y} - X\underline{\theta}\|^2}$$

likelihood ratio test for $H_0: \underline{\theta} = \underline{0}$ vs. $H_a: \underline{\theta} \neq \underline{0}$.

$$\Lambda = \frac{\text{max. likelihood under } H_0}{\text{max. likelihood under } H_0 \cup H_a}$$

we reject if the value of Λ is small

$$\Leftrightarrow \log \Lambda = \text{max log lik. under } H_0 - \text{max. log lik. under } H_0 \cup H_a$$

It turns out that the likelihood ratio test becomes equivalent to the F test.